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EXTENDED ABSTRACT

Interpreting SAE Features for Wildfire Forecasting

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1. Introduction & Objectives

In recent years, wildfire severity has increased globally due to climate change, making reliable and global approaches for monitoring more important than ever. At the same time, the rapid expansion of Earth observation data contributed to the emergence of deep learning-based wildfire forecasting models tackling those needs [1]. However, current forecasting models act as a black box compared to commonly used fire weather indices, limiting their adoption by wildfire experts in real-world applications. The field of eXplainable Artificial Intelligence (XAI) addresses this limitation by describing the decisions made by models using human-understandable concepts. Sparse Autoencoders (SAEs), a class of dictionary learning methods, have emerged as a concept-based approach to identify monosemantic features within the activations of a deep learning model by learning high-dimensional sparse representations [2].

In this work, we investigate the potential of SAEs to enhance human understanding of wildfire forecasting models on the SeasFire dataset [1]. The features of SAEs are usually interpreted with LLMs [2] in natural language tasks and with concept activation maps [4] in visual tasks. However, neither of these methods applies to SeasFire's environmental inputs [5], due to the lack of a vision-language model trained on this data, the complexity of the spatial patterns involved, and the interdependencies between the variables. To make SAE interpretation possible with SeasFire, we propose a set of rule-based meta-explainers interpreting high-dimensional SAE features into humanly understandable rules, see Figure 1.

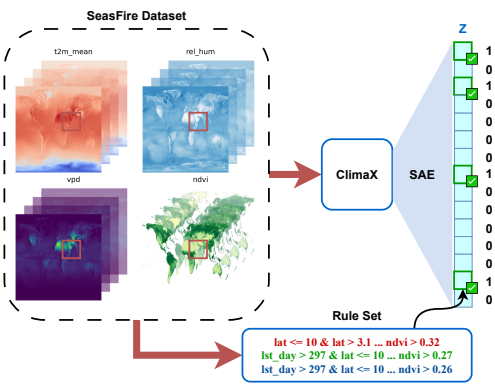

Figure 1. *Overview of our framework: Rule sets are trained to interpret the activations of each dimension in a SAE applied to ClimaX [3] and SeasFire [1].*

2. Methods & Data

SAE Training. We consider a deep learning model $f : \mathcal{X} \to \mathcal{Y}$, in which $f^{(l)} : \mathcal{X} \to \mathbb{R}^d$ is the mapping from the input $\mathcal{X}$ to the activations at layer (l). For a sample $x \in \mathcal{X}$, we define the vector $h = f^{(l)}(x) \in \mathbb{R}^d$ as the input of the SAE. We use a $k-$sparse autoencoder (TopKSAE [6]) to directly control the number of active features via the TopK function:

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$$
\begin{aligned}
z &= \text{TopK}(W_{enc}h + b_{enc}), \\
\hat{h} &= W_{dec}z,
\end{aligned}
\tag{1}
$$

with $W_{enc} \in \mathbb{R}^{n \times d}$, $b_{enc} \in \mathbb{R}^n$, and $W_{dec} \in \mathbb{R}^{d \times n}$ the parameters of our TopKSAE. The training loss is the MSE: $\mathcal{L} = \|h - \hat{h}\|_2^2$.

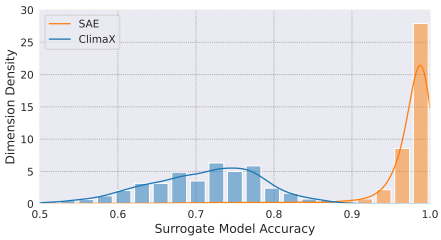
Rule-based Meta-Explainer. We rewrite Equation 1 as follows: $\hat{h} = \sum_{i=0}^{n-1} z_i d_i$, with z_i the activation of the feature d_i, a column of matrix W_{dec}. An emergent property of SAEs' high-dimensional and sparse latent space is the monosemanticity of its features [2]. As a consequence, we hypothesize that, for each feature d_i, we can learn a surrogate interpretable model $g_i : \mathcal{X} \to \{0, 1\}$ that maps the input space to the binarized activation of the feature d_i: $\tilde{z}_i(x) = \mathbb{1}[z_i(x) > 0]$. Each model g_i is defined as a finite set of rules $\mathcal{R}_i = \{r_{i,1}, ..., r_{i,m}\}$, where each rule $r_{i,j} : \mathcal{X} \to \{0, 1\}$ is a conjunction of threshold conditions on the input variables with a normalized weight w_r: $g_i(x) = \mathbb{1}[\sum_{r \in \mathcal{R}_i} w_r r(x) > 0.5]$. The training and evaluation sets of each model g_i, trained via Skope-rules, are sampled in a balanced way across $\mathcal{X}$, similarly to the LLM autointerpretability design [2].

Data. We finetuned ClimaX [3] on a set of 14 variables from the SeasFire dataset, on the 8-day horizon dense forecasting task [1]. As we are interpreting a dense task, the vectors $x \in \mathcal{X}$ and $h \in f^{(l)}(\mathcal{X})$ represent patches of 4×4 pixels at SeasFire native resolution of 0.25°.

3. Preliminary results

Model	Metric	Value
ClimaX	F1-score ↑	0.560
	PRAUC ↑	0.620
TopKSAE	R^2 ↑	0.927
	MSE ↓	1.84e-3
	Dead Codes ↓	0.159

(a) ClimaX and TopKSAE performance.



(b) Surrogate models accuracy on SAE and ClimaX activations.

Figure 2. *Model performance and interpretability analysis on SeasFire.*

In Table 2a, we present the performance of ClimaX on the dense fire forecast task (*rows 1-2*), and the reconstruction error of the TopKSAE (*row 3-5*) on the test set of SeasFire. TopKSAE is trained on the last layer of ClimaX before the classification head and presents strong reconstruction performance and utilization rate (i.e., a low percentage of dead codes). Figure 2b presents the initial results using the learned surrogate meta-explainers: the distribution of performance for the surrogate models trained to predict the binarized activations of the TopKSAE features, versus the threshold activations of the corresponding ClimaX layer (considered as a baseline). The difference in performance confirms our hypothesis: the TopKSAE features are more monosemantic than the neuron activations.

References

[1] I. Karasante, L. Alonso, I. Prapas, A. Ahuja, N. Carvalhais, and I. Papoutsis, "Seasfire cube-a multivariate dataset for global wildfire modeling," *Scientific Data*, vol. 12, no. 1, p. 368, 2025.

[2] R. Huben, H. Cunningham, L. Smith, A. Ewart, and L. Sharkey, "Sparse Autoencoders Find Highly Interpretable Features in Language Models," in *International Conference on Representation Learning*, 2024.

[3] T. Nguyen, J. Brandstetter, A. Kapoor, J. K. Gupta, and A. Grover, "ClimaX: A foundation model for weather and climate," in *International Conference on Machine Learning*, 2023.

[4] T. Fel, V. Boutin, L. Béthune, R. Cadene, M. Moayeri, L. Andéol, M. Chalvidal, and T. Serre, "A Holistic Approach to Unifying Automatic Concept Extraction and Concept Importance Estimation," in *Advances in Neural Information Processing Systems*, 2023.

[5] H. Porta, I. Kamoun, and D. Tuia, "Interpretable by-design wildfire forecasting via prototypes," in *EGU General Assembly Conference Abstracts*, 2025.

[6] A. Makhzani and B. Frey, "K-sparse autoencoders," *arXiv preprint arXiv:1312.5663*, 2013.